

Explore America

Interactive Multimedia Web Application

CS 430 – Internet Multimedia Programming

Professor Bryan Harley

Spring 2026

Wari L. Seattle Page – Multimedia Integration

Miguel B. San Juan Page – Design/CSS

Cole Snuffer Home Page & Trip Planner

May 2026

`waril2001.github.io/Explore-america-multimedia`

Contents

1	Project Overview	2
1.1	Live Website	2
2	Design Process	2
2.1	Planning	2
2.2	Visual Design	2
2.3	Structure	2
2.4	Accessibility	3
3	Wire-frames	3
3.1	Home Page	3
3.2	Seattle Page	4
3.3	San Juan Page	4
3.4	Trip Planner Page	5
4	Technologies Used	5
5	Key Features	5
5.1	Destination Selector	6
5.2	Seattle Page	6
5.3	San Juan Page	6
5.4	Trip Planner	6
5.5	Contact Form	6
5.6	About Page	6
5.7	Credits Page	6
5.8	Accessibility Features	7
6	Challenges Faced	7
6.1	Fulfilling All Requirements	7
6.2	JavaScript Logic	7
6.3	File Structure and Asset Management	7
6.4	Accessibility Implementation	7
6.5	GitHub Collaboration	7
7	How GitHub Was Used	8
7.1	Repository Setup	8
7.2	Branch-Based Workflow	8
7.3	Pull Requests and Code Review	8
7.4	Commit History	8
7.5	GitHub Pages Deployment	8
8	Individual Contributions	8
9	References	8

Project Overview

Explore America is an interactive multimedia web application developed as the final group project for CS 430 Internet Multimedia Programming. The goal of the project was to design and build a fully functional travel guide that helps users discover two iconic American destinations like Seattle, Washington, and San Juan, Puerto Rico. Based on their personal travel preferences.

The application allows users to select their preferred climate on the home page and get redirected to a destination page with rich multimedia content including videos, image galleries, and audio. It also includes an interactive trip planner where users can generate a personalized day-by-day itinerary based on their travel style, trip duration, budget, and interests.

The project integrates all major concepts covered in CS 430, including semantic HTML5, external CSS with responsive design, JavaScript interactivity, multimedia integration, accessibility, and ethical use of media assets.

Live Website

<https://waril2001.github.io/Explore-america-multimedia/>

Design Process

Planning

We started by breaking down the project requirements and identifying what each page needed to include. We divided the work according to the strengths of each member. Wari handled Seattle and multimedia integration, Miguel handled San Juan and visual design, and Cole handled the home page and trip planner logic.

Visual Design

We chose a dark navy blue (`#1e3a5f`) and orange (`#fca311`) color scheme that felt professional and travel-themed on all pages. Each destination page has its own visual identity. Seattle uses a hero image with an overlay card and a video section, while San Juan uses a full-page beach background with floating card elements and a video sec.

Structure

The site follows a clean multi-page structure with a consistent header, navigation bar, main content area, and footer across all pages. The JavaScript files were kept modular; `main.js` handles the shared logic across all pages, while `city.js`, `sanjuan.js`, `planner.js` and `contact.js` handle page-specific interactivity.

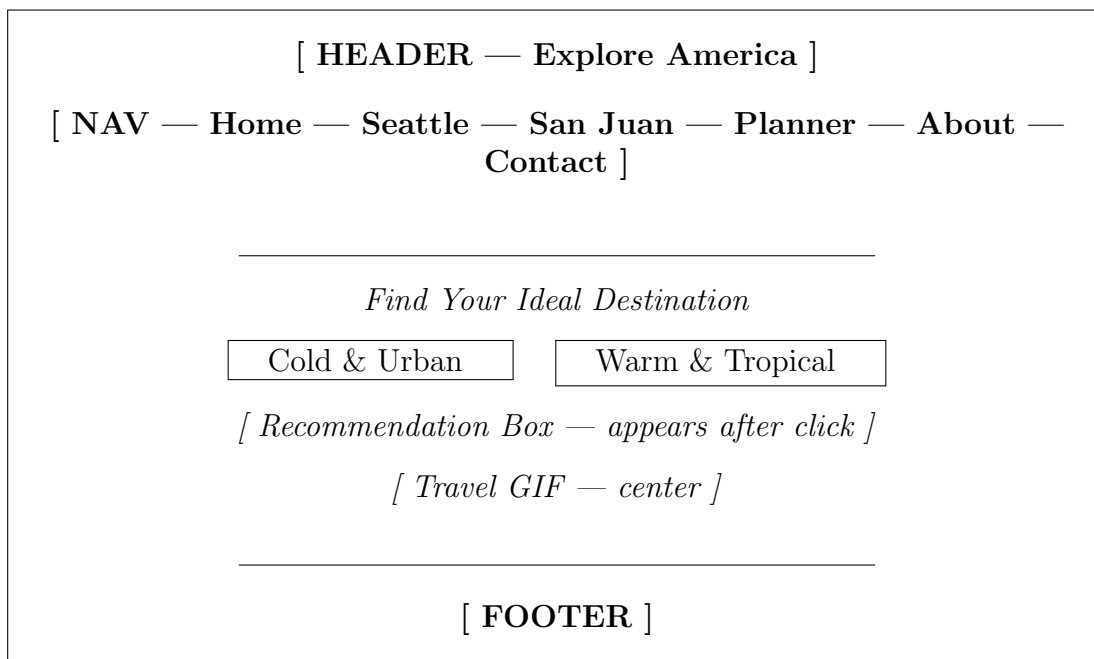
Accessibility

From the start we made accessibility a priority. Every page includes a skip-to-main-content link, `aria-label` attributes on navigation and media elements, `aria-live` regions for dynamic content, `role="alert"` on form error messages, descriptive alt text on all images, and visible keyboard focus styles throughout the site.

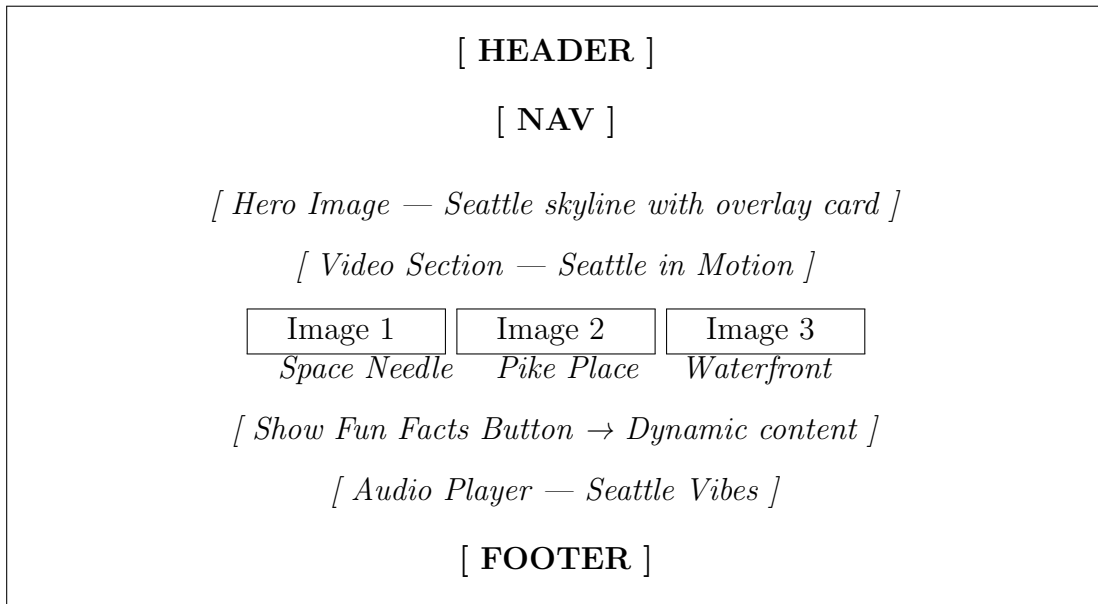
Wire-frames

The following wireframes illustrate the planned layout for each page before development began. Each page follows a consistent structure with a header, navigation bar, main content area, and footer.

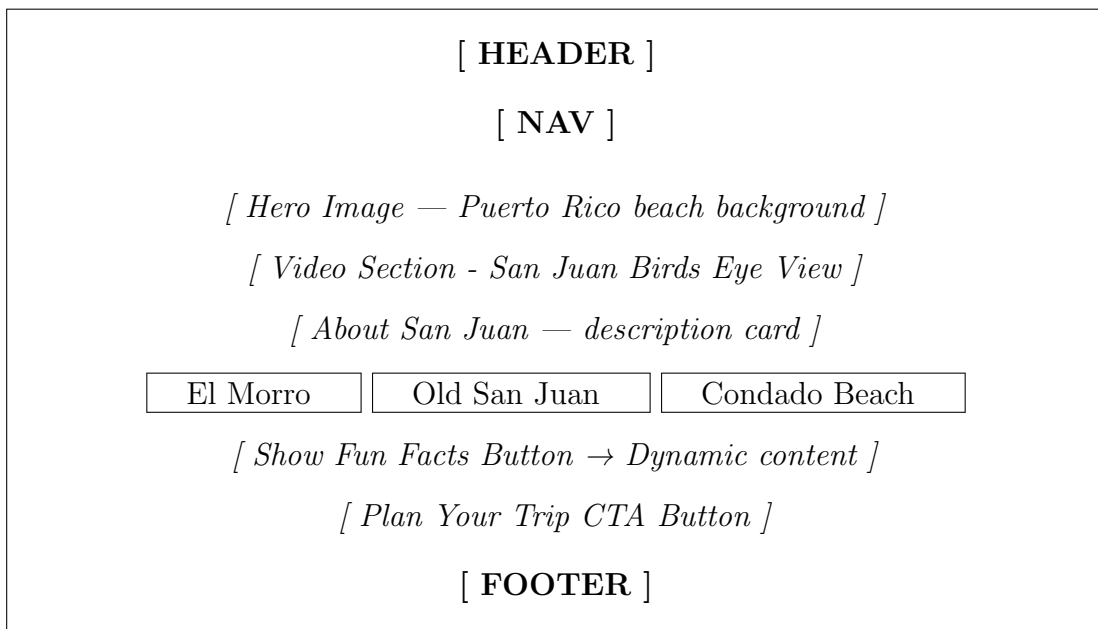
Home Page



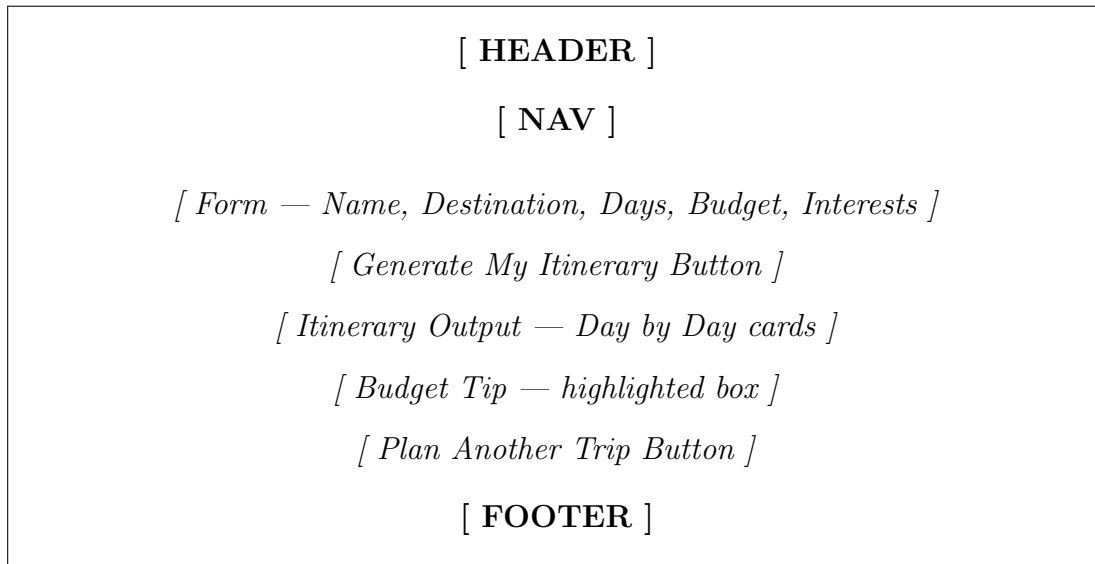
Seattle Page



San Juan Page



Trip Planner Page



Technologies Used

Technology	Purpose
HTML5	Semantic page structure using header, nav, main, section, article, and footer tags
CSS3	External stylesheet with Flexbox layout, animations, hover effects, media queries, and responsive design
JavaScript	DOM manipulation, event listeners, client-side form validation, and dynamic content generation
Lottie Animations	Animated travel GIF on the home page via LottieFiles (Lottie Simple License)
GitHub	Version control, branch-based collaboration, pull requests, and code review
GitHub Pages	Free static site hosting and deployment of the live website
Unsplash	Copyright-free photography for destination images (Unsplash License)

Table 1: Technologies and tools used in the Explore America project

Key Features

Destination Selector

The home page features two choices Cold & Urban and Warm & Tropical, which then routes users to their matching destination page. When clicked, a recommendation box appears with a brief description of the city before redirecting, giving users a preview of what to expect.

Seattle Page

The Seattle page features a full hero image with an overlay content card, a video tour of the city with playback controls and a text description fallback, and an image gallery with hover zoom effects showcasing the Space Needle, Pike Place Market, and the Seattle Waterfront. The page also includes a fun facts button that dynamically displays interactive content using JavaScript, along with an ambient audio player equipped with accessibility labels.

San Juan Page

The San Juan page features a beach background image, a hero section with an overlay card, and a brief overview highlighting the city's history and culture. It also includes an image gallery showcasing three popular attractions: El Morro, Old San Juan, and Condado Beach. A fun facts button provides interactive DOM content, while a call to action button directs users to the trip planner.

Trip Planner

The trip planner is the most interactive feature of the site. Users fill out a form with their name, destination, number of days (1–7), budget level, and travel interests. The form includes client-side validation with inline error messages for every required field. On successful submission, the planner generates a personalized day-by-day itinerary with two activities per day pulled from a curated list based on the user's selected interests and destination. A budget tip is also displayed at the bottom of the itinerary.

Contact Form

The contact page features a form with fields for name, email, subject, and message. Each field includes client side validation with inline error messages to ensure accurate user input. Upon successful submission, a thank you message is displayed and the form automatically resets.

About Page

The about page presents the three team members with their roles and contributions, along with a list of technologies used in the project.

Credits Page

The credits page features a complete attribution table documenting all media assets used throughout the project, including images, video, audio, animations, and fonts. Sources and

licensing information are clearly organized and properly credited.

Accessibility Features

Every page includes a skip to main content link, aria label attributes for navigation and media elements, aria live regions for dynamically generated content, role alert attributes for form error messages, descriptive alt text for all images, and visible keyboard focus styles defined in the external stylesheet.

Challenges Faced

Fulfilling All Requirements

One of the main challenges was ensuring that every project requirement was fully implemented, including semantic HTML, JavaScript interactivity, multimedia integration, accessibility, and the ethical use of media, while still keeping the code clean and well organized. This required carefully reviewing the project guidelines multiple times to ensure that no important details or features were overlooked.

JavaScript Logic

Building the trip planner was the most technically demanding part of the project. The form needed to validate multiple field types, handle checkbox groups, pull relevant activities based on the user's selected interests and destination, and generate a structured day-by-day output dynamically. Getting all of that to work together without errors took significant debugging and iteration.

File Structure and Asset Management

Early in the project, we maintained two separate folders for media assets, Assets and media, which led to inconsistent naming conventions and file organization. To improve structure and maintainability, we reorganized all assets into a single media folder with dedicated subfolders for images, video, audio, and animations. We also updated all file path references throughout the HTML and CSS files to ensure consistency across the project.

Accessibility Implementation

Adding proper accessibility features was more involved than expected. Beyond just adding alt text to images, we had to implement skip links, ARIA attributes, keyboard focus styles, and `aria-live` regions for dynamic content. Understanding how screen readers interact with dynamically injected content required research and testing.

GitHub Collaboration

Working with branches and pull requests was a new workflow for the team. We had to learn how to create feature branches, commit regularly with meaningful messages, open pull

requests, and request reviews before merging to `main`. There were a few instances where we had to coordinate carefully to avoid overwriting each other's work.

How GitHub Was Used

Repository Setup

One team member created the repository on GitHub with a public visibility setting, initialized with a `README.md` and a `.gitignore` file. The repository name follows the required format for the course.

Branch-Based Workflow

Each team member worked on their own feature branch rather than committing directly to `main`. Feature branches were named descriptively (e.g., `feature/about-contact-pages`) to make it clear what changes each branch contained.

Pull Requests and Code Review

All changes were merged into `main` through pull requests. Team members were added as reviewers before merging, following the collaboration rules established in the project guidelines.

Commit History

We committed regularly with meaningful commit messages that described what was changed and why. This made it easy to track the evolution of the project and identify when specific features were added.

GitHub Pages Deployment

We used GitHub Pages to deploy the live website directly from the `main` branch. The site is accessible at:

`https://waril2001.github.io/Explore-america-multimedia/`

Any changes merged into `main` are automatically reflected on the live site within a few minutes.

Individual Contributions

References

1. Klepacki, S. (n.d.). *Castillo San Felipe del Morro, San Juan, Puerto Rico* [Photograph]. Unsplash. https://unsplash.com/photos/zU41e_hj58g

Team Member	Contributions
Wari L.	Built the Seattle destination page including the hero image section, video tour with accessibility labels and fallback content, image gallery with hover zoom effects, fun facts button with dynamic DOM manipulation, and the ambient audio player. Also handled multimedia optimization and implemented accessibility features across the project including skip links, ARIA attributes, and keyboard focus styles.
Miguel B.	Built the San Juan destination page including the full-page beach background, hero overlay card, about section, image gallery with the three top attractions, fun facts button, and the call-to-action button linking to the planner. Also handled the visual design direction, color scheme, and CSS animations throughout the site.
Cole Snuffer	Built the home page including the destination selector buttons, recommendation box, and travel GIF integration. Developed the trip planner with full form validation, interest-based itinerary generation, budget tips, and reset functionality. Also built the contact form with client-side validation and success message handling. Established the overall navigation structure used across all pages.

Table 2: Individual contributions by team member

- George, A. (n.d.). *Colorful streets of Old San Juan with Puerto Rico flag* [Photograph]. Unsplash. <https://unsplash.com/photos/-JAwGkwd2UM>
- Whitcotton, L. (n.d.). *Condado lagoon with San Juan skyline and palm tree* [Photograph]. Unsplash. <https://unsplash.com/photos/v6vVfXjn4ao>
- MJ. (n.d.). *Puerto Rican flag on beach* [Photograph]. Unsplash. <https://unsplash.com/photos/fh6D0ab3ELE>
- MJ. (n.d.). *Bird's-eye view of San Juan, Puerto Rico* [Video]. Pexels. <https://www.pexels.com/video/37460317/>
- Unsplash. (n.d.). *Space Needle, Seattle* [Photograph]. Unsplash. <https://unsplash.com>
- Unsplash. (n.d.). *Pike Place Market, Seattle* [Photograph]. Unsplash. <https://unsplash.com>
- Unsplash. (n.d.). *Seattle Waterfront* [Photograph]. Unsplash. <https://unsplash.com>

com

9. Get Set Go!! (n.d.). *Free Travel Animation* [Lottie animation]. LottieFiles. <https://lottiefiles.com/free-animation/travel-UnHgld39R>
10. Mozilla Developer Network. (n.d.). *HTML: HyperText Markup Language*. MDN Web Docs. <https://developer.mozilla.org/en-US/docs/Web/HTML>
11. Mozilla Developer Network. (n.d.). *CSS: Cascading Style Sheets*. MDN Web Docs. <https://developer.mozilla.org/en-US/docs/Web/CSS>
12. Mozilla Developer Network. (n.d.). *JavaScript*. MDN Web Docs. <https://developer.mozilla.org/en-US/docs/Web/JavaScript>
13. GitHub. (n.d.). *GitHub Pages documentation*. <https://docs.github.com/en/pages>
14. LottieFiles. (n.d.). *Lottie Simple License*. <https://lottiefiles.com/page/license>
15. Unsplash. (n.d.). *Unsplash License*. <https://unsplash.com/license>